

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1605

Vasona 138 kV
March 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1605. The Interconnection Customer proposes a 746.5 MW natural gas combined cycle facility interconnecting at the Vasona 138 kV substation in the West Virginia area. The request was received on November 11, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 519 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$108,495,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of OAKELY / VASONA 138.0 kV, the VASONA / EXCELL 138.0 kV L loads to 375.9% of its 198 MVA rating. Project DFAX: 100.0%.

For the loss of EXCELL / VASONA 138.0 kV line, the VASONA / OAKELY 138.0 kV line loads to 368.0% of its 202 MVA rating. Project DFAX: 100.0%.

The OAKELY / W MURRAY SUBSTATION 138.0 kV Line is overloaded to 360.4% of its 206 MVA emergency rating for the loss of the VASONA / EXCELL 138.0 kV Line. The project contributes 100.0% DFAX (746.5 MW) to this overload.

For the loss of VASONA / EXCELL 138.0 kV, the TITANS / W MURRAY 138.0 kV loads to 355.0% of its 208 MVA rating. Project DFAX: 100.0%.

The VONORE / TITNS 138.0 kV L is overloaded to 312.6% of its 229 MVA emergency rating for the loss of the VASONA / EXCELL 138.0 kV. The project contributes 100.0% DFAX (746.5 MW) to this overload.

The COALINGA 1 / EXCELL 138.0 kV is overloaded to 278.3% of its 181 MVA emergency rating for the loss of the OAKELY / VASONA 138.0 kV Line. The project contributes 62.9% DFAX (469.4 MW) to this overload.

For the loss of VASONA / EXCELL 138.0 kV L, the VONORE 138/1kV Transformer loads to 221.6% of its 323 MVA rating. Project DFAX: 100.0%.

The VONORE 345 kV/1 kV Autotransformer is overloaded to 204.9% of its 349 MVA emergency rating for the loss of the VASONA / EXCELL 138.0 kV line. The project contributes 100.0% DFAX (746.5 MW) to this overload.

For the loss of COALINGA 1 SUBSTATION / EXCELL 138.0 kV, the EXCELL / CARPENTER 138.0 kV loads to 199.8% of its 175 MVA rating. Project DFAX: 42.9%.

The CARPENTER / LANDIS STATION 138.0 kV is overloaded to 191.8% of its 181 MVA emergency rating for the loss of the EXCELL / COALINGA 1 138.0 kV. The project contributes 42.9% DFAX (320.0 MW) to this overload.

The LANDIS STA SUBSTATION / DRESDEN DRIVE 138.0 kV is overloaded to 179.0% of its 191 MVA emergency rating for the loss of the EXCELL / COALINGA 1 138.0 kV. The project contributes 42.9% DFAX (320.0 MW) to this overload.

DRESDEN DRIVE / DELTA PERSON LLC 138.0 kV: 168.0% loading (200 MVA) following the outage of the EXCELL / COALINGA 1 138.0 kV; DFAX = 42.9%.

The MARKET / BOULDER CRK 138.0 kV is overloaded to 148.3% of its 202 MVA emergency rating for the loss of the COALINGA 1 / FITZWILLIAM 138.0 kV line. The project contributes 30.7% DFAX (229.3 MW) to this overload.

For the loss of FITZWILLIAM / COALINGA 1 138.0 kV line, the COALINGA 1 SUBSTATION / BOULDER CREEK 138.0 kV loads to 146.1% of its 210 MVA rating. Project DFAX: 30.7%.

FITZWILLIAM / COALINGA 1 138.0 kV Line: 131.3% loading (211 MVA) following the outage of the COALINGA 1 / BOULDER CREEK 138.0 kV; DFAX = 30.4%.

The FITZWILLIAM / GOLD BEACH 138.0 kV is overloaded to 120.2% of its 263 MVA emergency rating for the loss of the COALINGA 1 / BOULDER CREEK 138.0 kV. The project contributes 21.4% DFAX (159.6 MW) to this overload.

The GOLD BEACH / GREEN BLUFF 138.0 kV is overloaded to 117.1% of its 437 MVA emergency rating for the loss of the GREENLAND GAP SUBSTATION / SANTEE COOPER 345.0 kV. The project contributes 17.7% DFAX (131.8 MW) to this overload.

COMERIO 9703 345/138kV Autotransformer: 106.0% loading (351 MVA) following the outage of the GREENLAND GAP SUBSTATION / SANTEE COOPER 345.0 kV; DFAX = 12.4%.

For the loss of RIO DELL / ST. JOHN 138.0 kV, the INTERNATIONAL PAPER VALLIANT OK / KEARNEYVILLE 345.0 kV loads to 105.8% of its 1,675 MVA rating. Project DFAX: 7.2%.

The COMERIO 9703 / RONKONKOMA 138.0 kV is overloaded to 105.4% of its 353 MVA emergency rating for the loss of the SANTEE COOPER / GREENLAND GAP 345.0 kV line. The project contributes 12.4% DFAX (92.5 MW) to this overload.

The SHADY GROVE / RONKONKOMA 138.0 kV is overloaded to 104.8% of its 369 MVA emergency rating for the loss of the GREENLAND GAP / SANTEE COOPER 345.0 kV. The project contributes 12.4% DFAX (92.5 MW) to this overload.

The SHADY GROVE / GREEN BLUFF 138.0 kV L is overloaded to 104.1% of its 385 MVA emergency rating for the loss of the SANTEE COOPER / GREENLAND GAP 345.0 kV. The project contributes 12.4% DFAX (92.5 MW) to this overload.

The AUURN RD SUBSTATION / PRIDES CORNER 345.0 kV is overloaded to 100.4% of its 1,662 MVA emergency rating for the loss of the RONCO / FIRSTENERGY FT MARTIN POWER STATION 345.0 kV L. The project contributes 30.7% DFAX (229.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EXCELL / VASONA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,078,000	100% (\$2,078,000)
VASONA / OAKELY 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,564,000	100% (\$1,564,000)
OAKELY / W MURRAY SUBSTATION 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,371,000	100% (\$6,371,000)
TITANS / WEST MURRAY 138.0 kV	Rebuild line with higher-capacity conductor	\$2,548,000	100% (\$2,548,000)
VONORE / TITNS 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,661,000	100% (\$3,661,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COALINGA 1 / EXCLL 138.0 kV	Rebuild line with higher-capacity conductor	\$4,448,000	100% (\$4,448,000)
VONORE 138/1kV Transformer	Replace transformer with higher-rated unit	\$5,792,000	100% (\$5,792,000)
VONORE 345 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$17,875,000	100% (\$17,875,000)
EXCELL / CARPENTER 138.0 kV	Rebuild line with higher-capacity conductor	\$2,954,000	96% (\$2,842,000)
CARPENTER / LANDIS STATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,624,000	96% (\$2,525,000)
LANDIS STA SUBSTATION / DRESDEN DRIVE 138.0 kV	Rebuild line with higher-capacity conductor	\$6,583,000	96% (\$6,333,000)
DRESDEN DRIVE / DELTA PERSON LLC 138.0 kV	Rebuild line with higher-capacity conductor	\$6,922,000	96% (\$6,659,000)
MARKET / BOULDER CRK 138.0 kV	Rebuild line with higher-capacity conductor	\$3,832,000	100% (\$3,832,000)
COALINGA 1 SUBSTATION / BOULDER CREEK 138.0 kV	Rebuild line with higher-capacity conductor	\$1,987,000	100% (\$1,987,000)
FITZWILLIAM / COALINGA 1 138.0 kV L	Rebuild line with higher-capacity conductor	\$9,529,000	100% (\$9,529,000)
FITZWILLIAM / GOLD BEACH 138.0 kV	Rebuild line with higher-capacity conductor	\$9,104,000	89% (\$8,140,000)
GOLD BEACH / GREEN BLUFF 138.0 kV	Rebuild line with higher-capacity conductor	\$3,222,000	77% (\$2,469,000)
COMERIO 9703 SUBSTATION 345/138kV XFMR	Replace transformer with higher-rated unit	\$15,347,000	76% (\$11,708,000)
INTERNATIONAL PAPER VALLIANT OK / KEARNEYVILLE 345.0 kV	Reconductor line	\$18,096,000	16% (\$2,934,000)
RONKONKOMA / COMRIO 9703 138.0 kV line	Reconductor line	\$995,000	76% (\$759,000)
SHADY GROVE / RONKONKOMA 138.0 kV	Reconductor line	\$3,726,000	76% (\$2,842,000)
SHADY GROVE / GREEN BLUFF 138.0 kV line	Reconductor line	\$1,288,000	76% (\$982,000)
PRIDES CORNER / AUBURN ROAD 345.0 kV line	Reconductor line	\$3,814,000	16% (\$617,000)
	Total allocated to project		\$108,495,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.